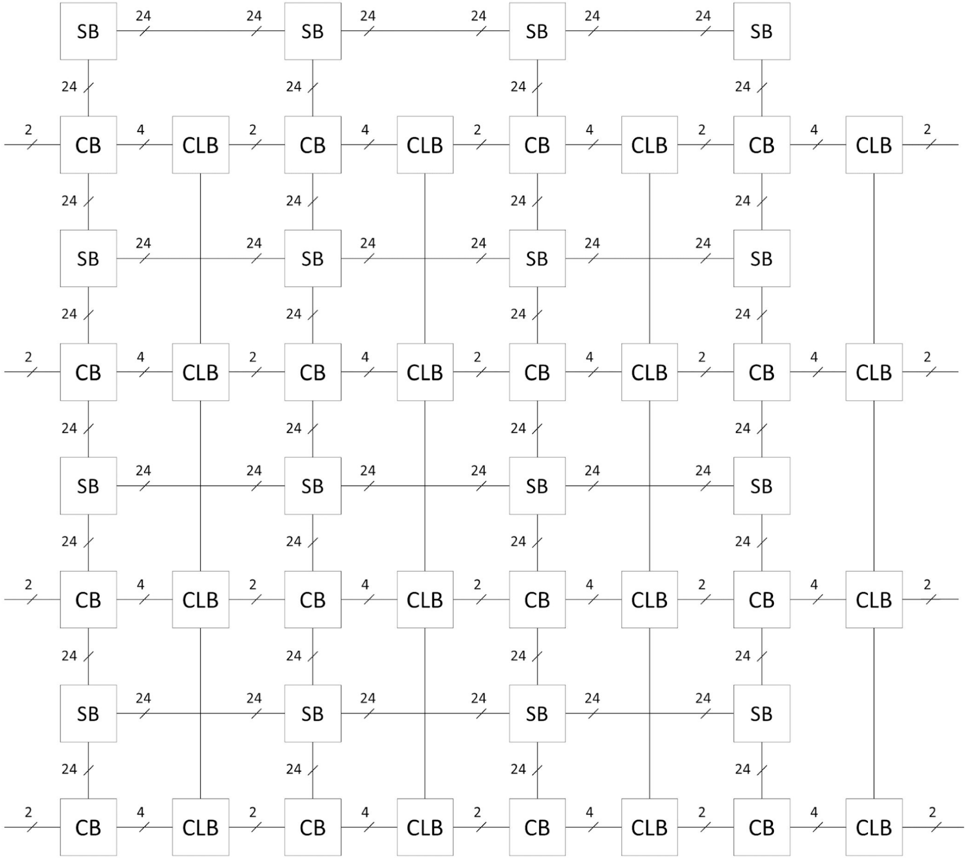
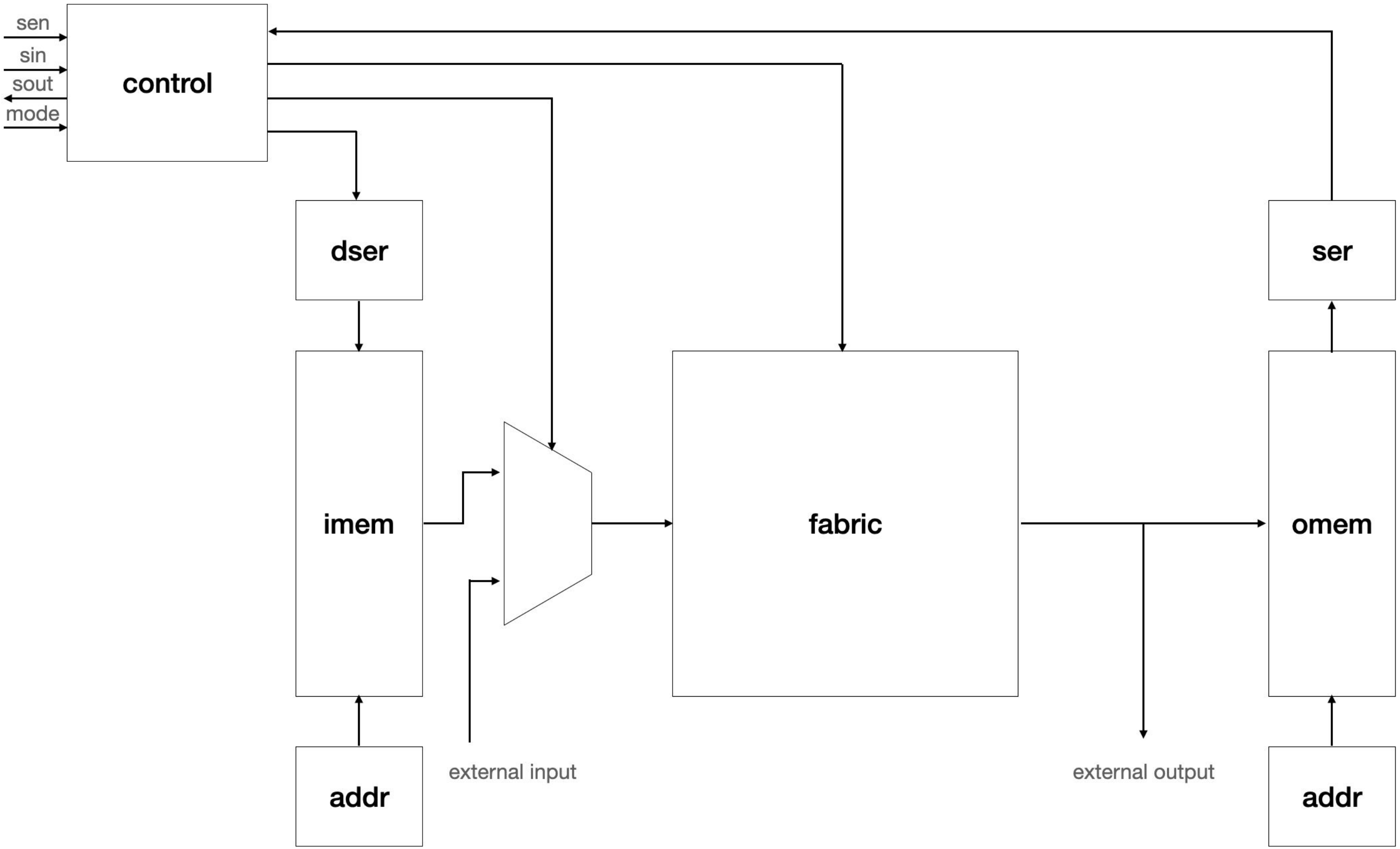
FPGA--

Max Lutwak, Shashank Obla, Nathan Serafin

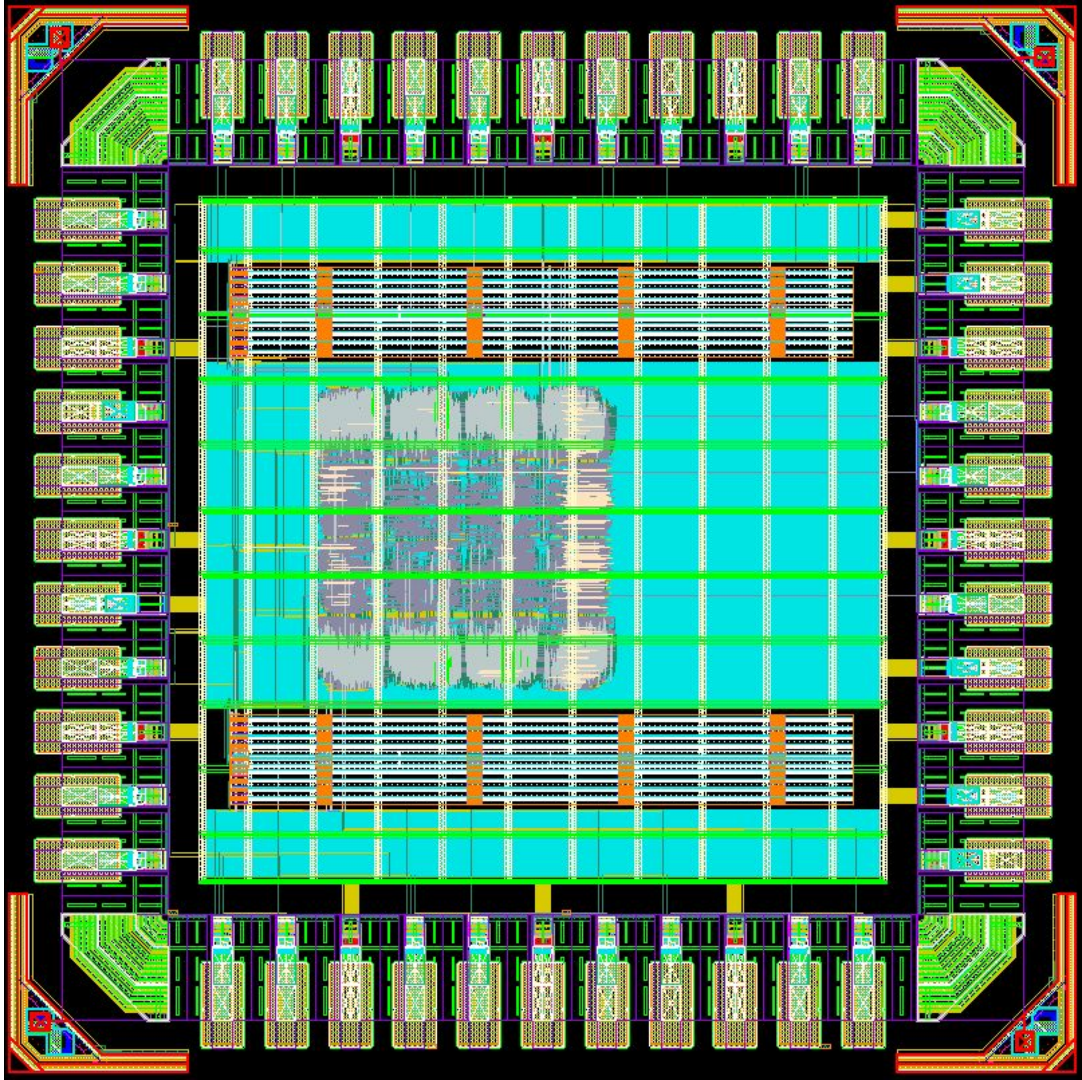
Fabric


SB 24 24 SB 24 24 SB 24 24 SB
24 24 24 24
2 CB 4 CLB 2 CB 4 CLB 2 CB 4 CLB 2 CB 4 CLB 2
24 24 24 24
SB 24 24 SB 24 24 SB 24 24 SB
24 24 24 24
2 CB 4 CLB 2 CB 4 CLB 2 CB 4 CLB 2 CB 4 CLB 2
24 24 24 24
SB 24 24 SB 24 24 SB 24 24 SB
24 24 24 24
2 CB 4 CLB 2 CB 4 CLB 2 CB 4 CLB 2 CB 4 CLB 2
24 24 24 24
SB 24 24 SB 24 24 SB 24 24 SB
24 24 24 24
2 CB 4 CLB 2 CB 4 CLB 2 CB 4 CLB 2 CB 4 CLB 2

Chip


sen
sin
sout
mode
control
dser
imem
addr
external input
fabric
external output
ser
omem
addr

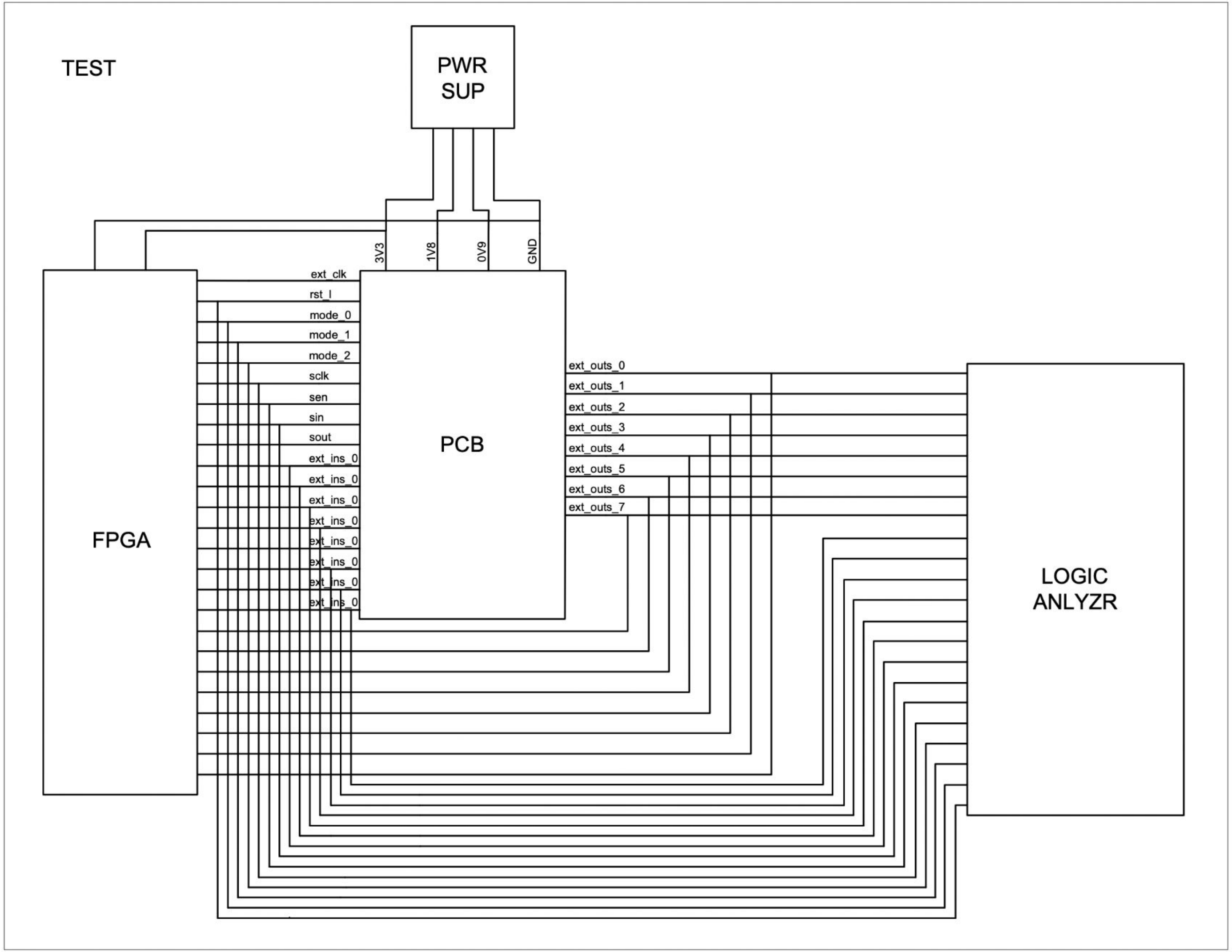
Layout



Design Specifications

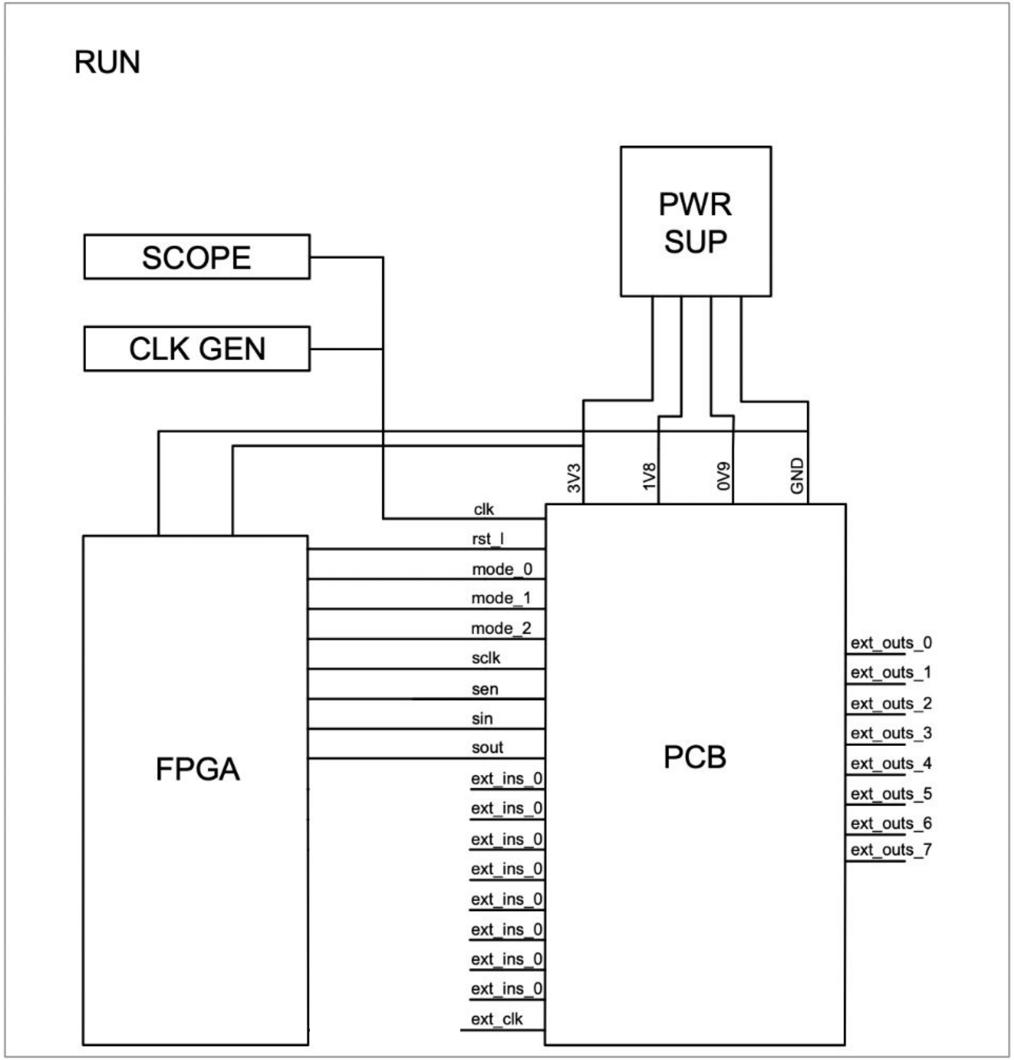
Clock Frequency	250MHz (worst-case bitstream)
Power	8.5mW
Area	1mm^2

Test Rig

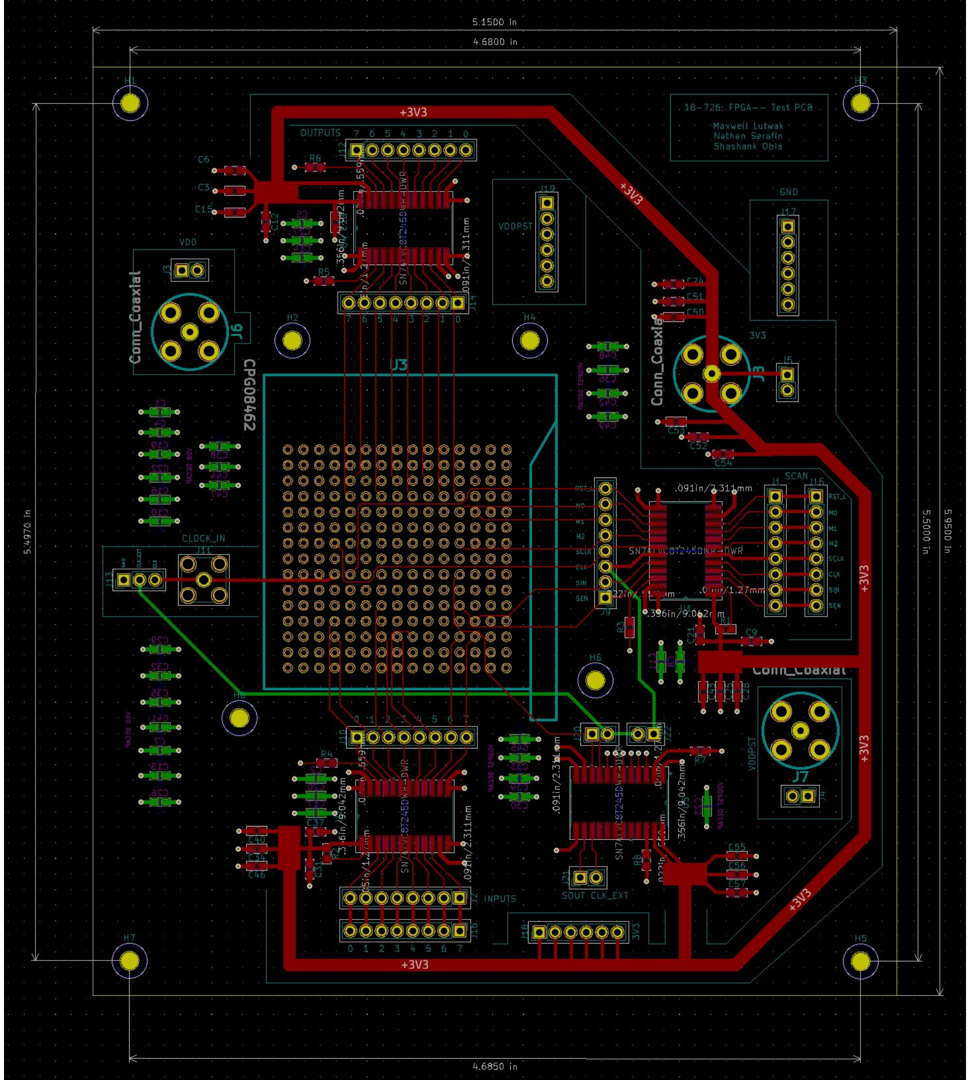


TEST
PWR
SUP
3V3
1V8
0V9
GND
FPGA
PCB
LOGIC
ANLYZR
ext_clk
rst_l
mode_0
mode_1
mode_2
sclk
sen
sin
sout
ext_ins_0
ext_ins_0
ext_ins_0
ext_ins_0
ext_ins_0
ext_ins_0
ext_ins_0
ext_ins_0
ext_outs_0
ext_outs_1
ext_outs_2
ext_outs_3
ext_outs_4
ext_outs_5
ext_outs_6
ext_outs_7

Test Rig

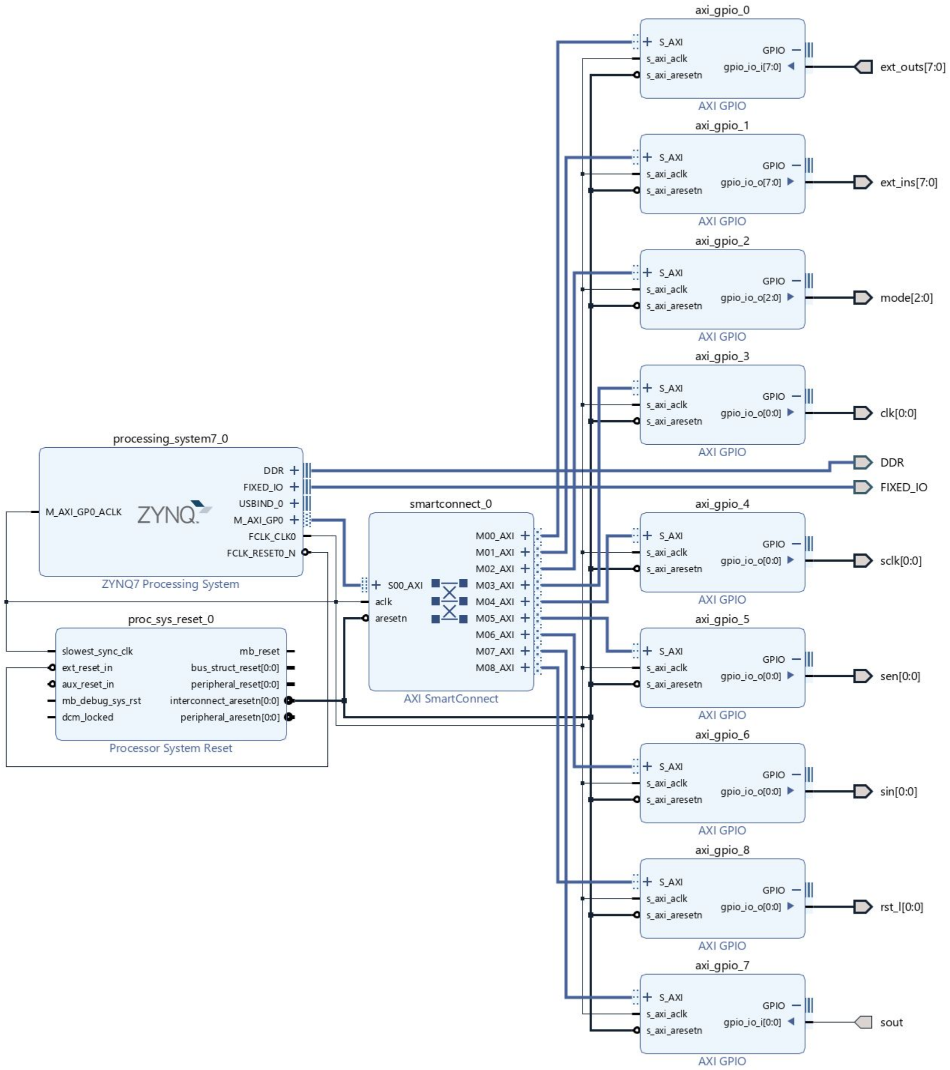

RUN
SCOPE
CLK GEN
PWR SUP
3V3
1V8
0V9
GND
clk
rst_l
mode_0
mode_1
mode_2
sclk
sen
sin
sout
ext_ins_0
ext_ins_0
ext_ins_0
ext_ins_0
ext_ins_0
ext_ins_0
ext_ins_0
ext_ins_0
ext_clk
FPGA
PCB
ext_outs_0
ext_outs_1
ext_outs_2
ext_outs_3
ext_outs_4
ext_outs_5
ext_outs_6
ext_outs_7

PCB

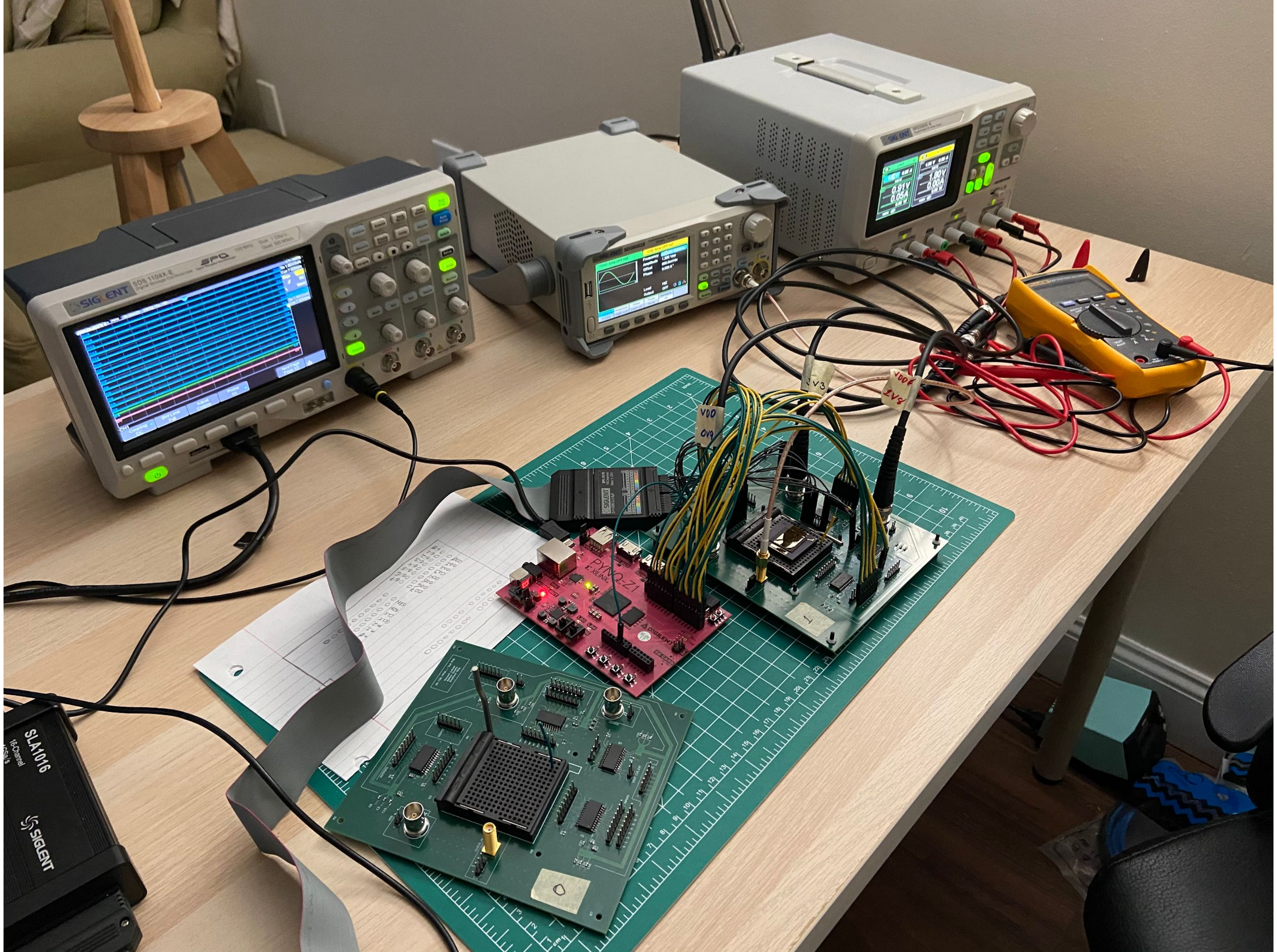


5.1500 in
4.6800 in
18-726: FPGA-- Test PCB
Maxwell Lutwak
Nathan Serafin
Shashank Obla
+3V3
OUTPUTS
VDD
Conn_Coaxial
CPG08462
J3
VDDPST
GND
3V3
SCAN
CLOCK_IN
INPUTS
SOUT CLK_EXT
5.4970 in
5.5000 in
5.9500 in
4.6850 in

FPGA Block Design



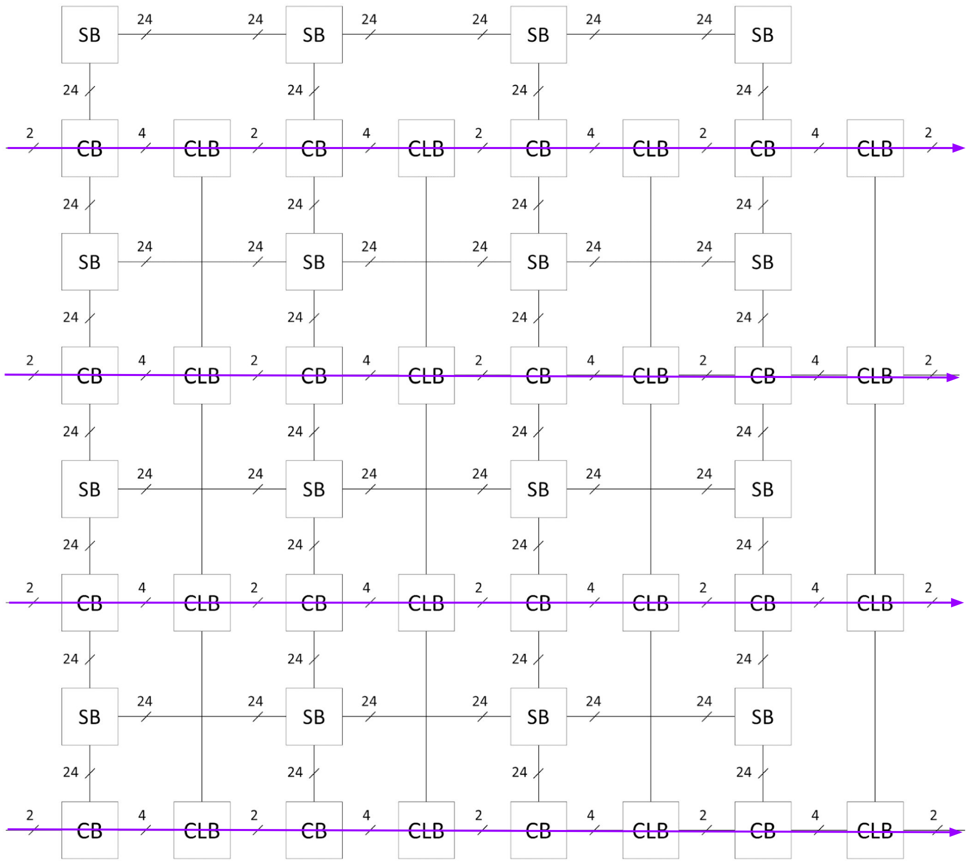
Test Setup


SIGLENT
SDS 1104X-E
PYNQ-Z1
SLA1016
16-Channel
SIGLENT
VDD
0V9
1V3
VDDE
2V8

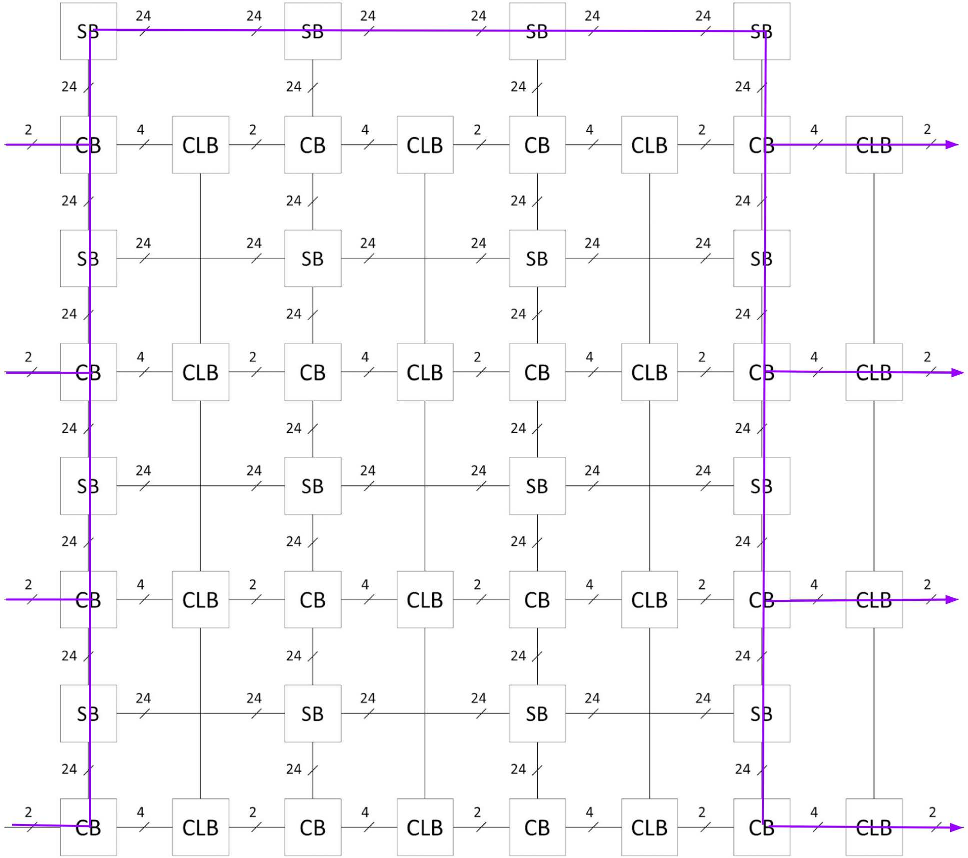
Bitstreams

Name	Function	Flops	Switch Blocks	Max Frequency
ones	1	N	N	N/A
identity	x	N	N	N/A
seq_identity	D4(x)	Y	N	500MHz
add_once	(x+y)	N	N	N/A
add	8*(x+y)	N	N	N/A
add_seq	D4(8*(x+y))	Y	N	N/A
sb_identity	x	N	Y	500MHz

Sequential Identity



SB Identity



Voltage vs. Frequency for SB Identity

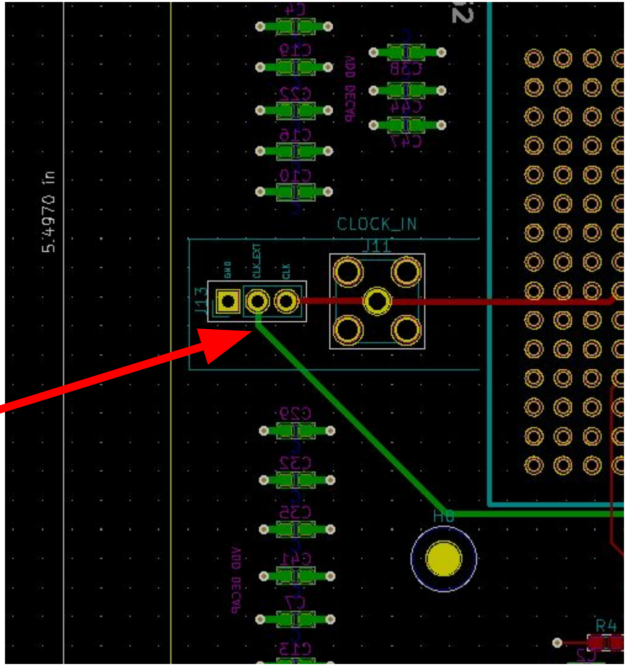
	100MHz	150MHz	200MHz	250MHz	300MHz	350MHz
0.55V	n	n	n	n	n	n
0.60V	y	y	y	n	n	n
0.65V	y	y	y	y	n	n
0.70V	y	y	y	y	y	n
0.75V	y	n	y	y	y	y
0.8V	y	y	y	y	y	y
0.85V	y	y	y	y	y	y
0.9V	y	y	y	y	y	y

Power

Measured (mW)	Simulated (mW)
45.5	4.9

Mostly board overhead? Turning the clock on has no measured effect.

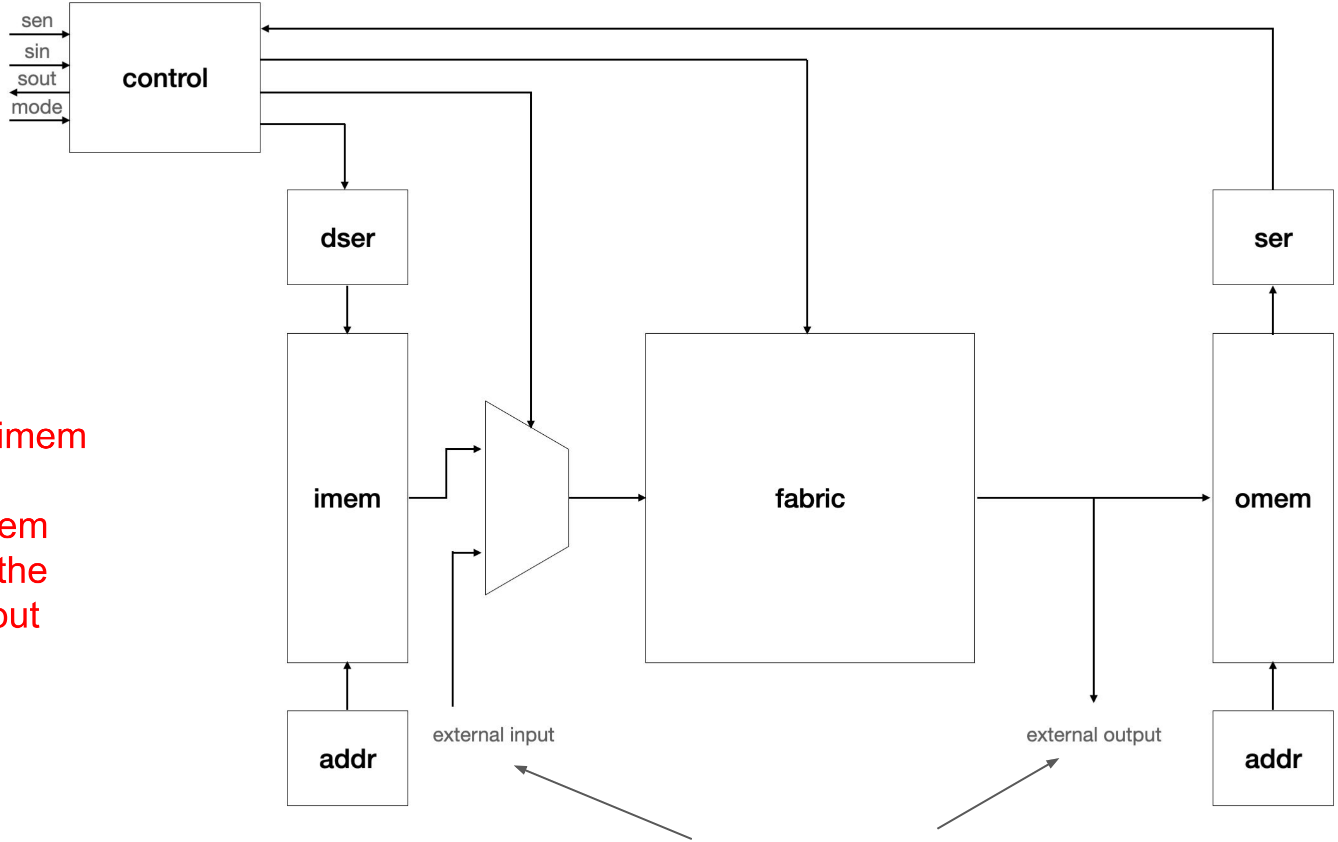
Mistakes



Jumper to disconnect long trace from high-speed clock

Makes it *impossible* to automate test sweeps

Mistakes



No visibility into imem and omem.
You *must* run imem vectors through the fabric and read out omem.

We have two 8-bit buses on chip outputs. A few extra muxes and we could have had full visibility, plus bytewise read/write.

Questions?